I. $x^3 + a_1 x^2 + a_2 x + a_3 = 0$

$$x^3 + 3\cdot\frac{a_1}{3}x^2 + 3\cdot\frac{a_1^2}{9}x + \frac{a_1^3}{27} - \frac{a_1^2}{3}x - \frac{a_1^3}{27} + a_2 x + a_3 = 0$$

$$\left(x + \frac{a_1}{3}\right)^3 - \frac{a_1^2}{3}x - \frac{a_1^3}{27} + a_2 x + a_3 = 0$$

$$z = x + \frac{a_1}{3}$$

$$z^3 - \frac{a_1^2}{3}\cdot\left(z - \frac{a_1}{3}\right) - \frac{a_1^3}{27} + a_2\left(z - \frac{a_1}{3}\right) + a_3 = 0$$

$$z^3 - \frac{a_1^2}{3}z + \frac{a_1^3}{9} - \frac{a_1^3}{27} + a_2 z - \frac{a_1 a_2}{3} + a_3 = 0$$

$$z^3 + z\cdot\left(a_2 - \frac{a_1^2}{3}\right) + a_3 + \frac{2a_1^3}{27} - \frac{a_1 a_2}{3} = 0$$

$$z^3 + az + b = 0$$

II. a) $b^2 + \frac{4a^3}{27} \geqslant 0$:

$$z = u + v$$

$$u^3 + 3u^2 v + 3uv^2 + v^3 + au + av + b = 0$$

$$u^3 + v^3 + 3uv\cdot(u+v) + a(u+v) + b = 0$$

$$\begin{cases} u^3 + v^3 = -b \\ u^3 v^3 = -\frac{a^3}{27} \end{cases} \Rightarrow t^2 + bt - \frac{a^3}{27} = 0 \Rightarrow D = b^2 + \frac{4a^3}{27}$$

$$t = \frac{-b \pm \sqrt{b^2 + \frac{4a^3}{27}}}{2} = -\frac{b}{2} \pm \sqrt{\frac{b^2}{4} + \frac{a^3}{27}}$$

$$z = \sqrt[3]{-\frac{b}{2} + \sqrt{\frac{b^2}{4} + \frac{a^3}{27}}} + \sqrt[3]{-\frac{b}{2} - \sqrt{\frac{b^2}{4} + \frac{a^3}{27}}}$$

$$x = \sqrt[3]{-\frac{b}{2} + \sqrt{\frac{b^2}{4} + \frac{a^3}{27}}} + \sqrt[3]{-\frac{b}{2} - \sqrt{\frac{b^2}{4} + \frac{a^3}{27}}} - \frac{a_1}{3}$$

б) $b^2 + \frac{4a^3}{27} < 0$:

$$\underline{4t^3 - 3t = A} \Rightarrow t^3 - \frac{3}{4}t = \frac{A}{4}$$

$$z = kt$$

$$k^3 t^3 + akt + b = 0$$

$$t^3 + \frac{at}{k^2} + b = 0 \Rightarrow k = 2\sqrt{-a/3}$$

$$4t^3 - 3t = A$$

$$t = \cos\alpha$$

$$4\cos^3\alpha - 3\cos\alpha = A \Rightarrow \cos 3\alpha = A \Rightarrow \alpha = \frac{\arccos A}{3} + \frac{2\pi k}{3},\ k \in \mathbb{Z}$$

$$\left[\begin{array}{l} t = \cos\left(\frac{1}{3}\arccos A\right) \Rightarrow z = 2\sqrt{-3/a}\,\cos\left(\frac{1}{3}\arccos A\right) \Rightarrow x = 2\sqrt{-3/a}\,\cos\left(\frac{1}{3}\arccos A\right) - \frac{a_1}{3} \\ t = \cos\left(\frac{1}{3}\arccos A + \frac{2\pi}{3}\right) \Rightarrow z = 2\sqrt{-3/a}\,\cos\left(\frac{1}{3}\arccos A + \frac{2\pi}{3}\right) \\ t = \cos\left(\frac{1}{3}\arccos A + \frac{2\pi}{3}\right) \Rightarrow z = 2\sqrt{-3/a}\,\cos\left(\frac{1}{3}\arccos A + \frac{2\pi}{3}\right) \end{array}\right.$$

а) $x^3+x^2-x+1=0$

$$a = a_2 - \frac{a_1^2}{3} = -1 - \frac{1}{3} = -\frac{4}{3}$$

$$b = a_3 + \frac{2a_1^3}{27} - \frac{a_1 a_2}{3} = 1 + \frac{2}{27} + \frac{1}{3} = \frac{36}{27} + \frac{2}{27} = \frac{38}{27}$$

$$z = x + \frac{a_1}{3} = x + \frac{1}{3}$$

$$z^3 - \frac{4}{3}z + \frac{38}{27} = 0$$

$$t = \frac{-19 \pm 3\sqrt{33}}{27}$$

$$z = \sqrt[3]{\frac{-19+3\sqrt{33}}{27}} + \sqrt[3]{\frac{-19-3\sqrt{33}}{27}}$$

$$x = \sqrt[3]{\frac{-19+3\sqrt{33}}{27}} + \sqrt[3]{\frac{-19-3\sqrt{33}}{27}} - \frac{1}{3}$$

б) $x^3+x^2-4x-4=0$

$$a = -\frac{13}{3}$$

$$b = -\frac{70}{27}$$

$$z = x + \frac{1}{3}$$

$$z^3 - \frac{13}{3}z - \frac{70}{27} = 0$$

$$z = kt = \frac{2\sqrt{13}}{3}t$$

$$\frac{8\cdot 13\cdot\sqrt{13}}{\cancel{27}}t^3 - \frac{13}{\cancel{3}}\cdot\frac{2\sqrt{13}}{\cancel{3}}t - \frac{70}{\cancel{27}} = 0$$

$$\frac{8\cdot 13\cdot\sqrt{13}}{3}t^3 - 13\cdot 2\sqrt{13}\,t - \frac{70}{3} = 0$$

$$8t^3 - 6t - \frac{70}{13\sqrt{13}} = 0$$

$$4t^3 - 3t = \frac{35}{13\sqrt{13}}$$

$$t = \cos\alpha$$

$$\cos 3\alpha = \frac{35}{13\sqrt{13}} \Rightarrow \alpha = \frac{1}{3}\arccos\frac{35}{13\sqrt{13}} + \frac{2\pi k}{3},\ k\in\mathbb{Z}$$

$$\left[\begin{array}{l} t = \cos\left(\frac{1}{3}\arccos\frac{35}{13\sqrt{13}}\right) \\ t = \cos\left(\frac{1}{3}\arccos\frac{35}{13\sqrt{13}} + \frac{2\pi}{3}\right) \\ t = \cos\left(\frac{1}{3}\arccos\frac{35}{13\sqrt{13}} - \frac{2\pi}{3}\right) \end{array}\right. \Rightarrow \left[\begin{array}{l} x = \frac{2\sqrt{13}}{3}\cos\left(\frac{1}{3}\arccos\frac{35}{13\sqrt{13}}\right) - \frac{1}{3} = 2 \\ x = \frac{2\sqrt{13}}{3}\cos\left(\frac{1}{3}\arccos\frac{35}{13\sqrt{13}} + \frac{2\pi}{3}\right) - \frac{1}{3} = -2 \\ x = \frac{2\sqrt{13}}{3}\cos\left(\frac{1}{3}\arccos\frac{35}{13\sqrt{13}} - \frac{2\pi}{3}\right) - \frac{1}{3} = -1 \end{array}\right.$$